IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 9: Edge Detection (10 Minutes)

Tasks:

1. Define Edge Detection in one sentence.

2. Name any two edge detection operators.

3. Which operator gives thin and accurate edges?

4. Why is Gaussian smoothing used before edge detection?

Answer:

1- Edge detection is the process of identifying sharp changes in intensity or brightness in an image to locate the boundaries of objects.

2- Sobel Operator and Prewitt Operator.

3- Canny Edge Detector.

4- Gaussian smoothing is used to reduce noise and unwanted variations in an image before detecting edges, which helps prevent false edges and improves the accuracy of edge detection.